

 Eskom	Physical Architecture Definition for a Physical Application Component	Group IT: EA
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Title: **PAD PAC for Remote
Temperature Logger**

Unique Identifier:

Project Identifier: **DEM-2788828**

Area of Applicability: **Eskom**

Documentation Type: **SP**

Disclosure
Classification: **CONTROLLED DISCLOSURE**

Project Stage Review

Lifecycle Phase	Lifecycle Stage	Stage Gate (after stage)	Deliverable	Appropriate Lifecycle Stage
Pre-Project Planning	Define Need		Project Statement of Work	
	Project Context	CRA	Statement of Architecture Work	
Concept	Feasibility			
	Concept Design	DRA	Conceptual Architecture Definition	
Definition	Logical (Basic) Design	ERA	Logical Architecture Definition	
Execution	Physical (Detailed) Design	IRA	Detailed Design	✓
Finalization	Pre-Transfer Design	FRA	Pre-Transfer Design	

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Enterprise Architecture
Manager**

Date: Date: Date: Date:

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DOCUMENT HISTORY

Revision No	Date	Author/s	Description
0.1	17 July 2024	Tawanda Chicheko	Draft
0.3	20 November 2024	Rajesh Ranjan	Draft
0.4	14 March 2025	Tawanda Chicheko	Draft
0.5	20 March	Johann Marx	Updates to draft version
0.6	3 April 2025	Tawanda Chicheko	Updates to final version
0.7	5 May 2025	Tawanda Chicheko	Security Architecture Updates

1 INTRODUCTION

1.1 PROJECT OVERVIEW

In 2015 a temperature logging system for the detection of overloaded transformers was piloted in KZN OU. The system was piloted as an improvement to thermal stickers which are an official Eskom recommended method of detecting overloaded transformers. However, as more thermal stickers are installed on transformer tanks, more effort is required to keep the transformer temperatures up to date as this requires CNC staff or Plant Management area staff to make frequent site visits to record or verify tank temperatures. The remote temperature logging system resolves these challenges by automating the data collection and analysis thus eliminating frequent transformer visits and hours spent analysing the collected data.

1.2 DOCUMENT SCOPE/PURPOSE

The Physical Architecture Definition (PAD)/Detailed Design contains content that is on a physical level of detail where it is needed for development purposes of the RTL Application.

The PAD spans all relevant architecture domains (business, data & information, application and technology) and examines all relevant states of the architecture (baseline, transition, and target).

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2 SUPPORTING CLAUSES

2.1 SCOPE

This document scope covers the following:

Applicability, normative/informative references, definitions, abbreviations and supporting documentation of this document.

Scope and context of architecture development work

Architecture designs for the physical application component

2.2 APPLICABILITY

This document shall apply throughout Eskom Holdings Limited Divisions.

2.3 NORMATIVE / INFORMATIVE REFERENCES

2.3.1 Normative

Ref #	Referenced Document	Description
1	Functional Specification Document	Digital Incubator- Remote Temperature Logger v1

2.3.2 Informative

Ref #	Referenced Document	Description
1		

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3 BUSINESS ARCHITECTURE

3.1 BUSINESS STAKEHOLDERS

Role	Name
Business Owner	Distribution (DX)
Business Requester	Peter Adigun

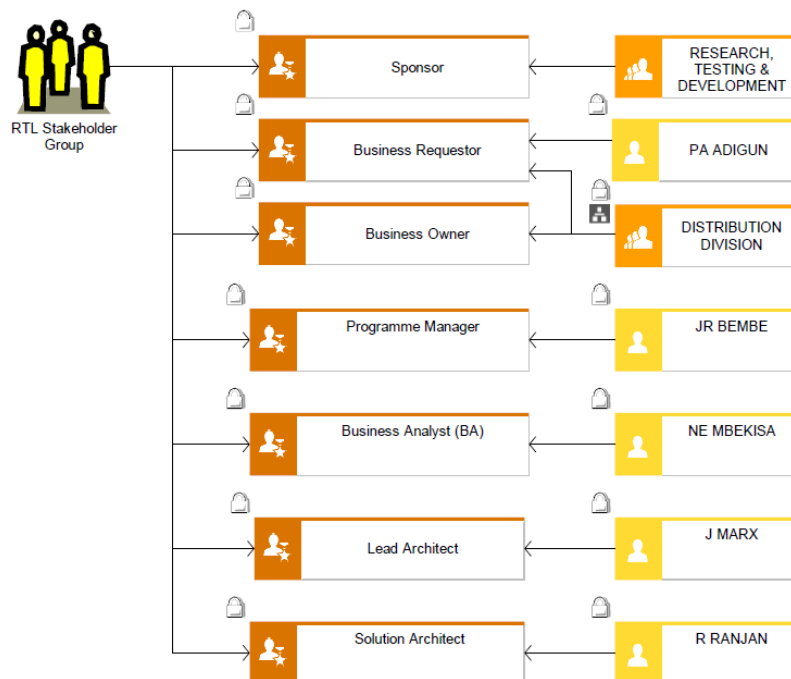
3.2 PROJECT STAKEHOLDER

Role	Name
Team Lead	Jabulani Bembe
Subject Matter Expert	Peter Adigun
Business Analyst	Ntsikelelo Mbekisa
Lead Architect	Johann Marx
Solution Architect	Tawanda Chicheko / Rajesh Ranjan

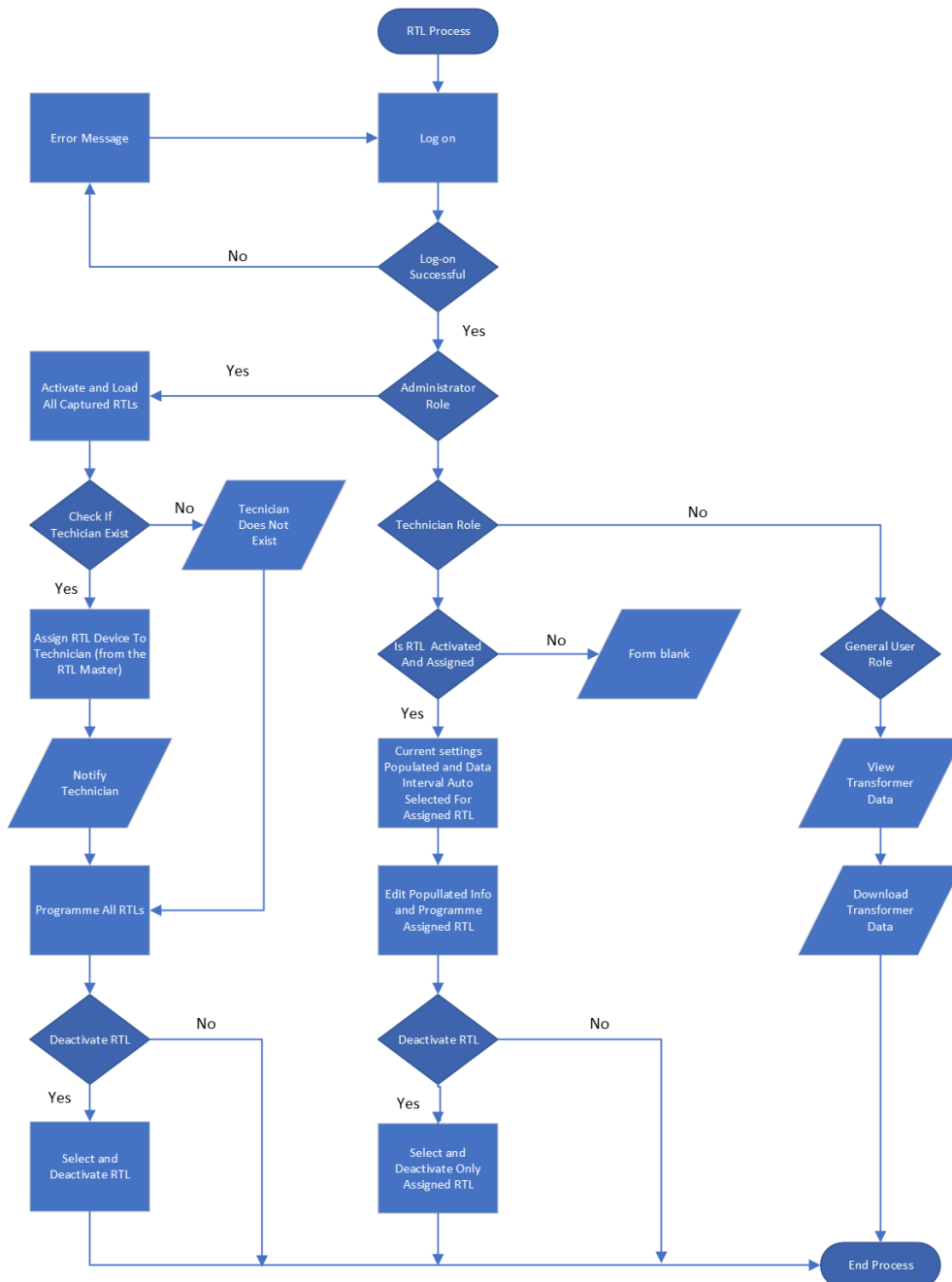
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3.3 TARGET STAKEHOLDER ANALYSIS DIAGRAM

Name: DEV_D1_Remote Temperature Logger Master_Target_202409_V02 Type: Stakeholder analysis diagram Identifier: STD.27965420	QA Date: 27 Jun 2014, 15:21:53 QA Status: Passed	Creator: chichet Time of generation: 27 Sep 2024, 12:21:20 Last user: ranjanr Last change: 13 Nov 2024, 15:17:52	 ARIS Model
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3.4 PROCESS FLOW



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3.5 TARGET APPLICATION USER LOCATIONS

The Pilot solution should cater for all Operating Units within Eskom's Distribution Division.

4 BUSINESS INFORMATION ARCHITECTURE

4.1 BUSINESS INFORMATION FLOW & INVENTORY

4.1.1 Business Information Flow Diagram

The Data flow from multiple applications to RTL Application and vice versa is displayed below.

The following Diagram has relevance:

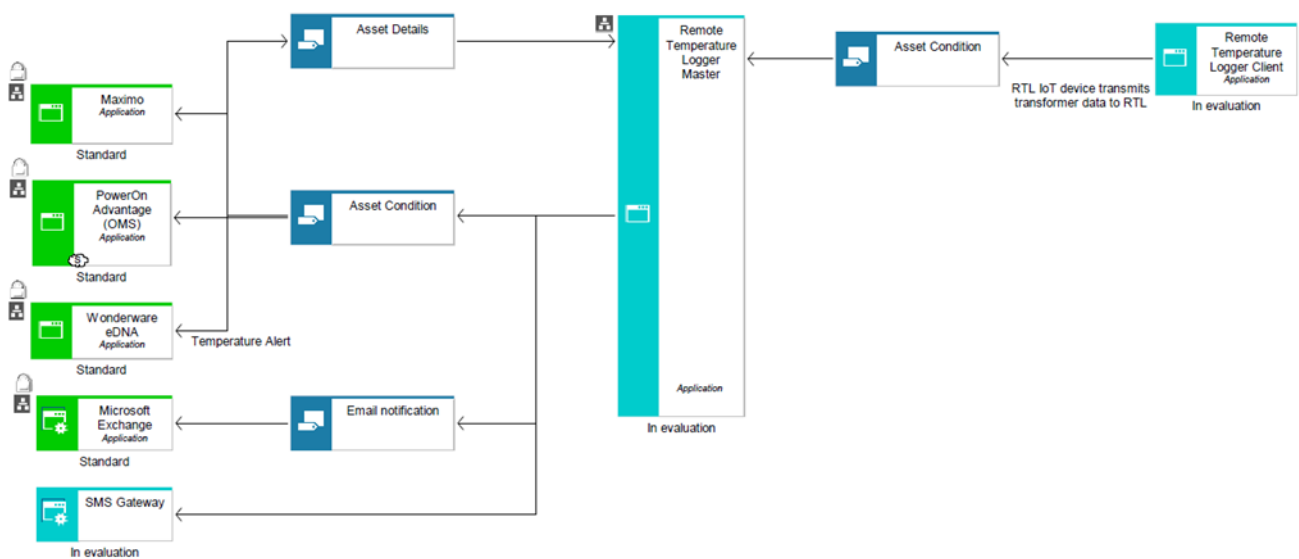


Figure: Data Flow Diagram

4.2 INFORMATION RELATED BUSINESS CONTINUITY

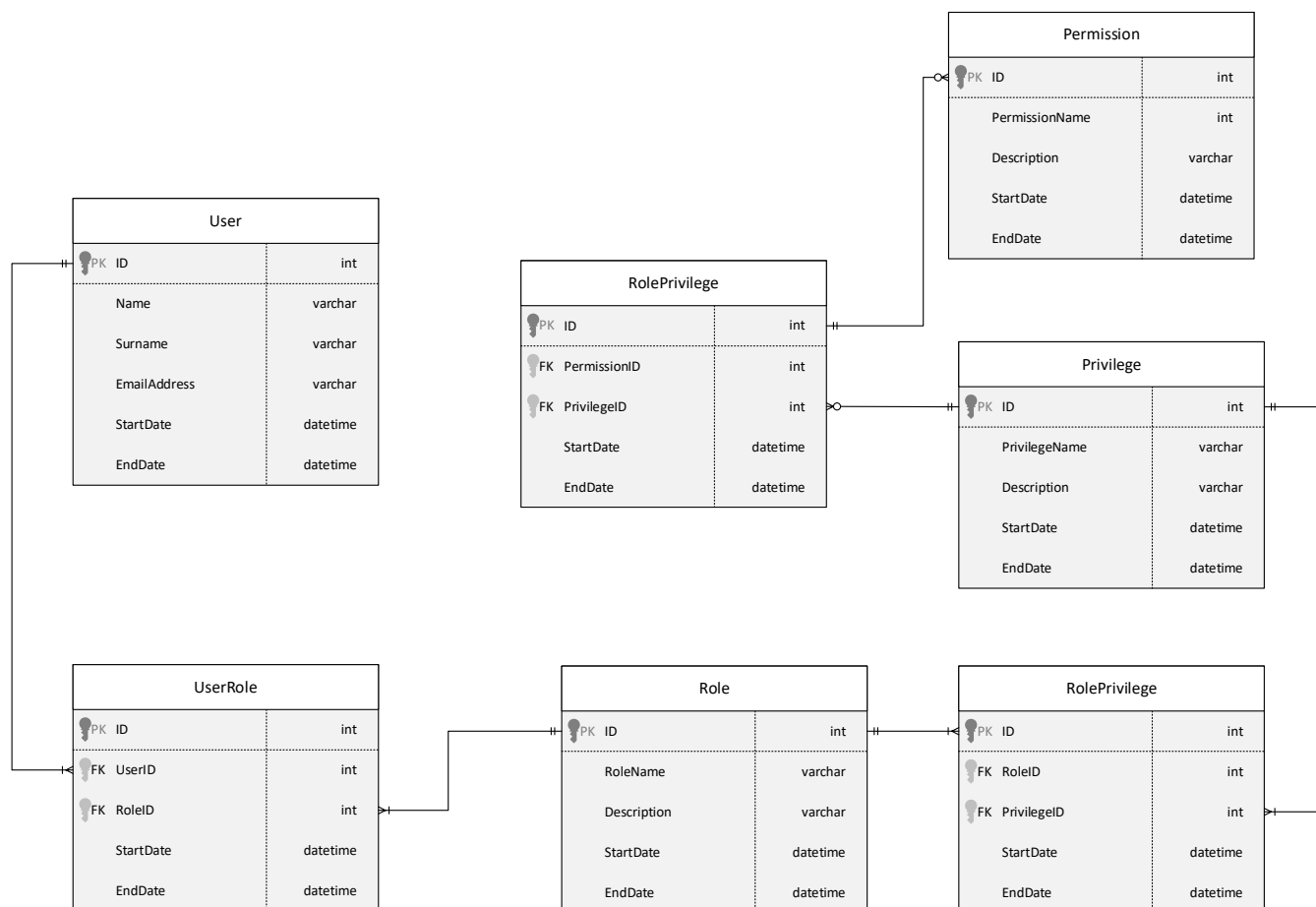
The BIA was not done for this is a Pilot initiative

5

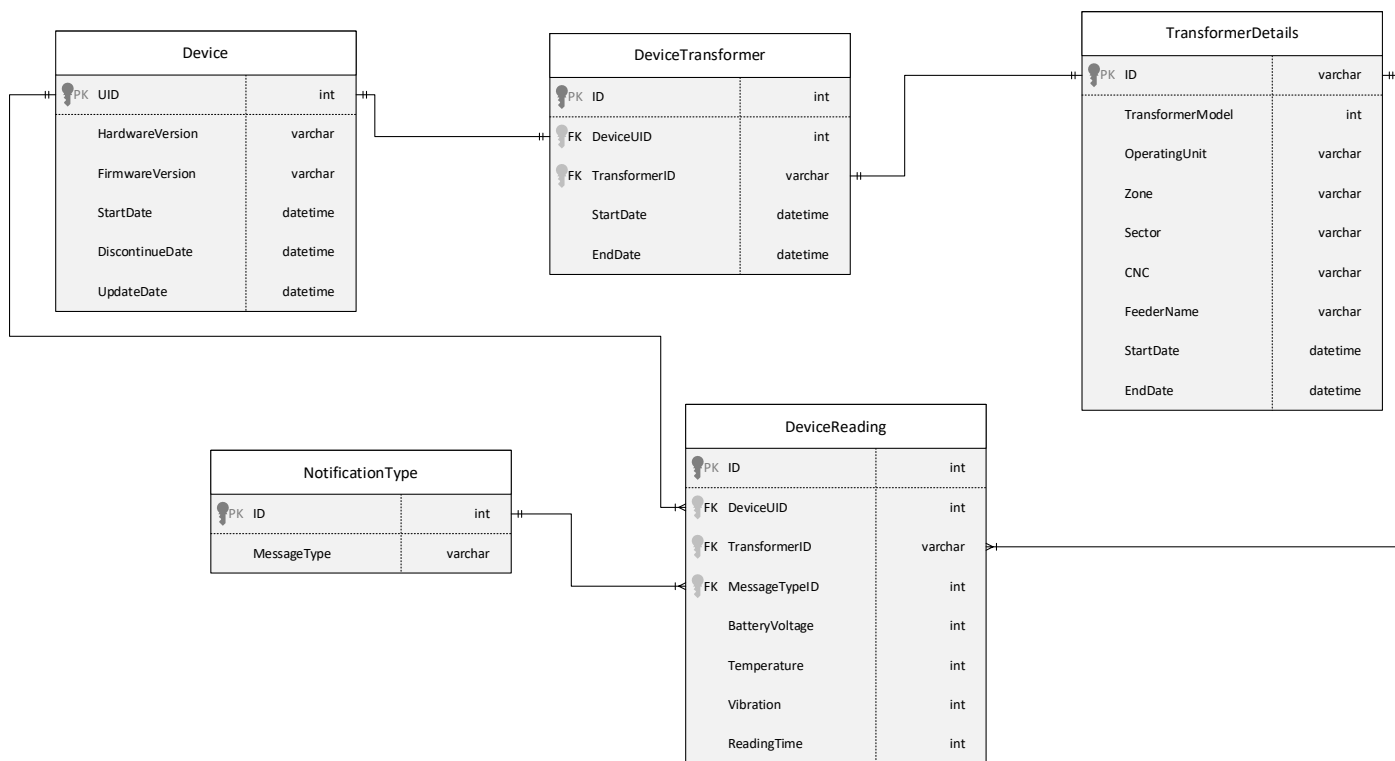
5 DATA ARCHITECTURE

5.1 DATA MODELS

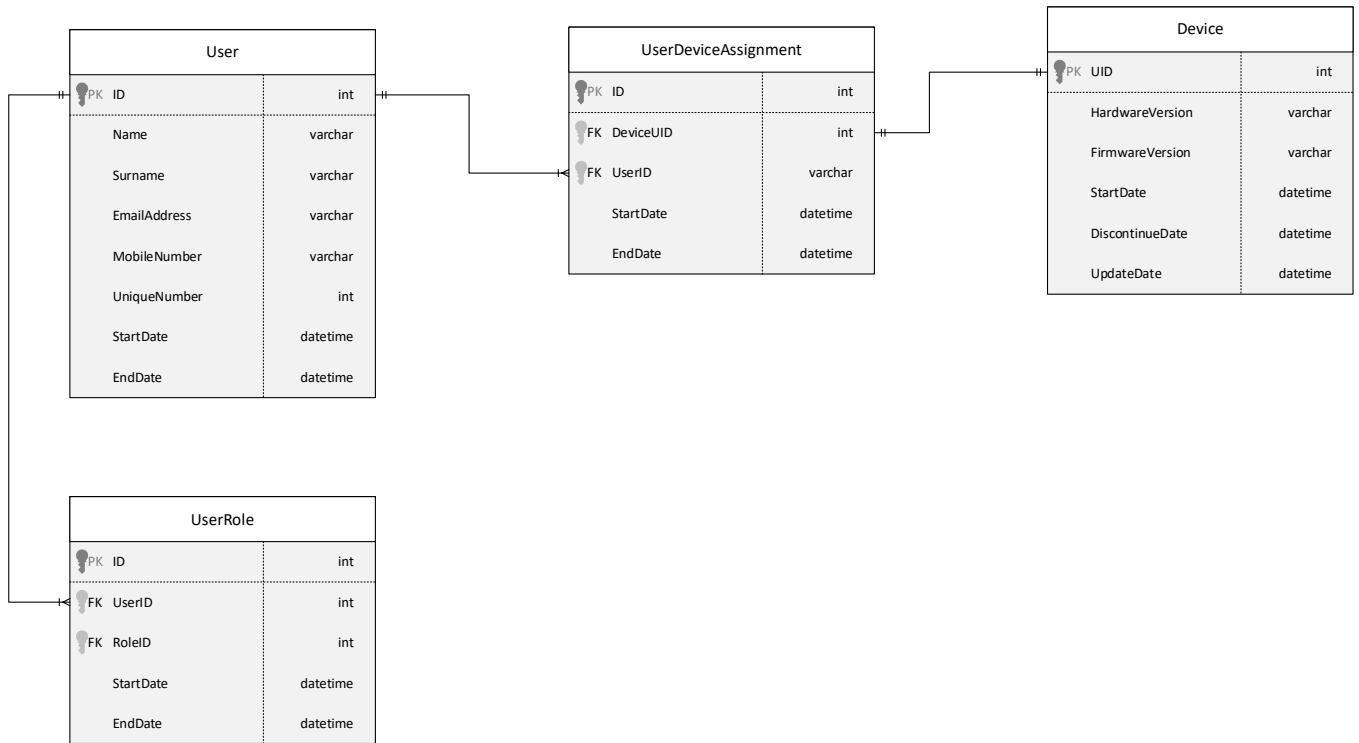
User Management Entity Relationship Diagram



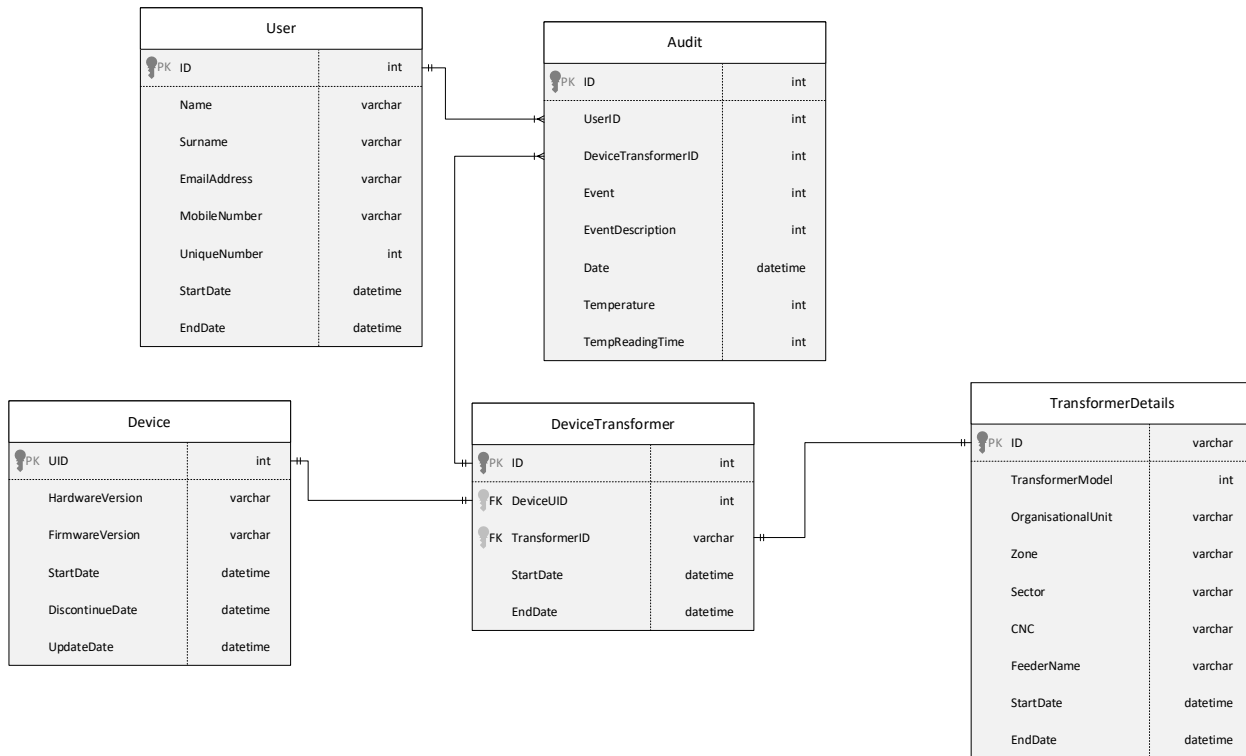
Transformer Temperature & Vibration Reading Entity Relationship Diagram



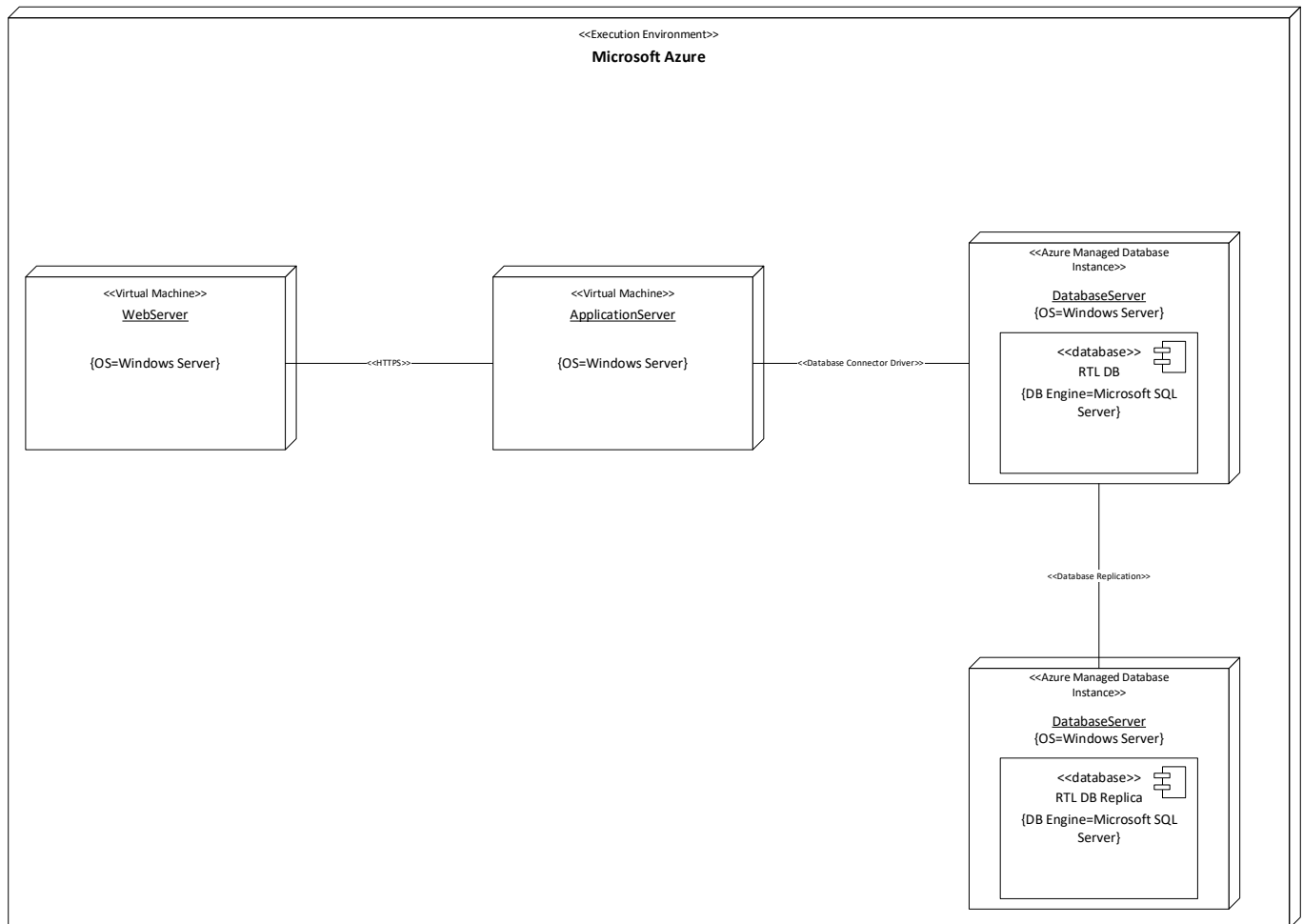
User- Device Assignment Entity Relationship Diagram



Audit Trail Entity Relationship Diagram



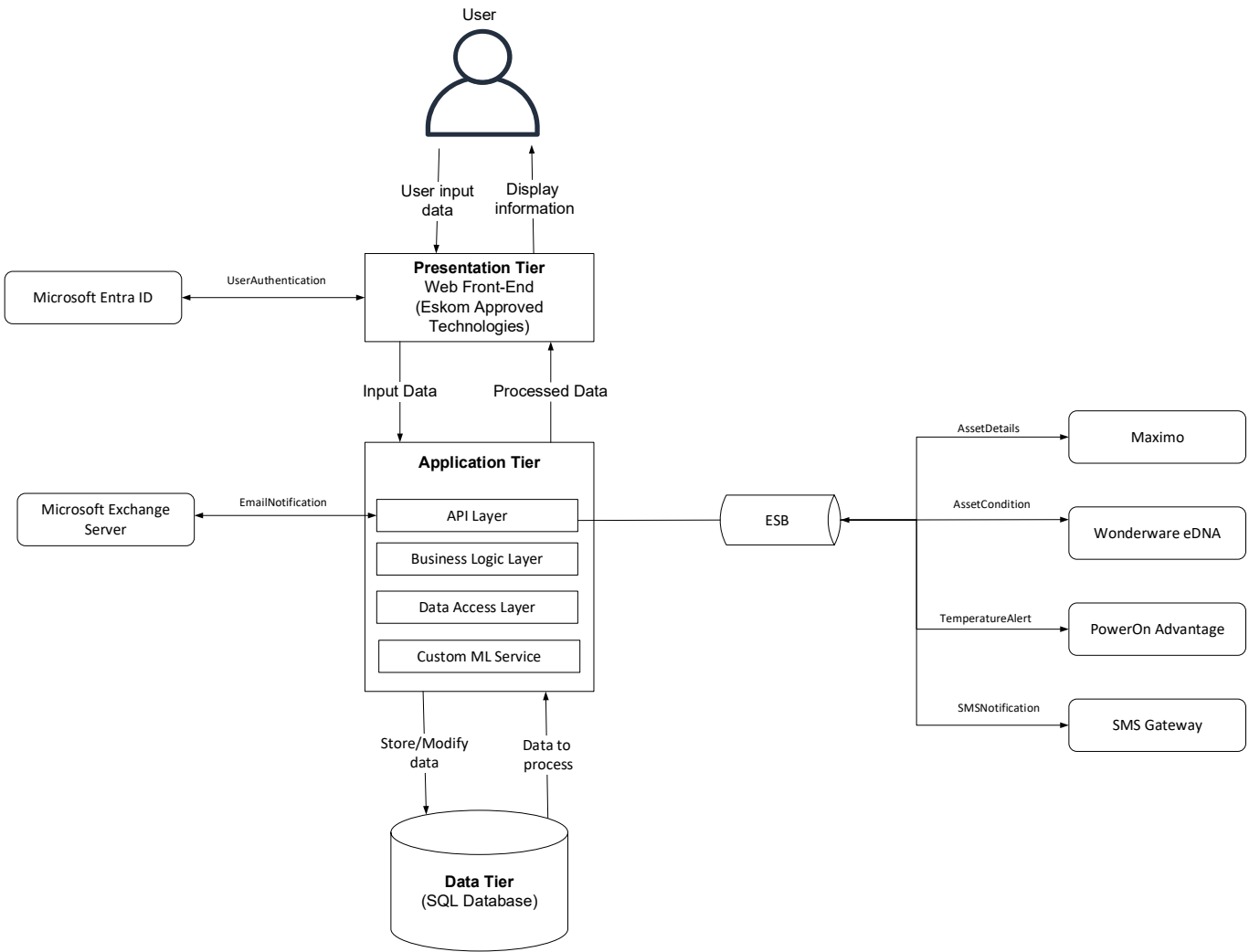
5.2 DATABASE DEPLOYMENT DIAGRAM



5.3 DATA MIGRATION

None

6 APPLICATION ARCHITECTURE



The RTL Application will make use of the 3-tier architecture consisting of the Presentation tier, Application tier and Data tier. This tiered architecture ensures efficient handling of requests and data, facilitating a robust and scalable application infrastructure.

Tier	Description
Presentation Tier	This purpose of this tier is to display information to and collect data from users. It provides the user interface of the application.
Application Tier	The Application Tier orchestrates the application's business logic, processing information from the Presentation Tier and interacting with the Data Tier for data manipulation. It is structured into three distinct layers for enhanced clarity and responsibility

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	segregation: the API layer acts as an interface, managing incoming requests and directing them to the appropriate business logic objects; the Business Logic layer processes these requests; and the Data Access layer executes SQL commands to manage database operations
Data Tier	This is the tier that handles the storage of data. Microsoft SQL Server database will be used.

6.1 SYSTEMS AFFECTED BY THIS IMPLEMENTATION

System	Impact
Maximo	Integrated to send asset details to the RTL Application. Integrated to receive asset condition information from RTL Application.
Wonderware eDNA	Integrated to receive asset condition information from RTL Application
PowerOn Advantage	Integrated to receive temperature alerts from RTL Application

6.2 SYSTEM USE BY THE BUSINESS

The solution is in Pilot stage and will only be used by the rest of Distribution Division business users once the pilot has been proven to be successful.

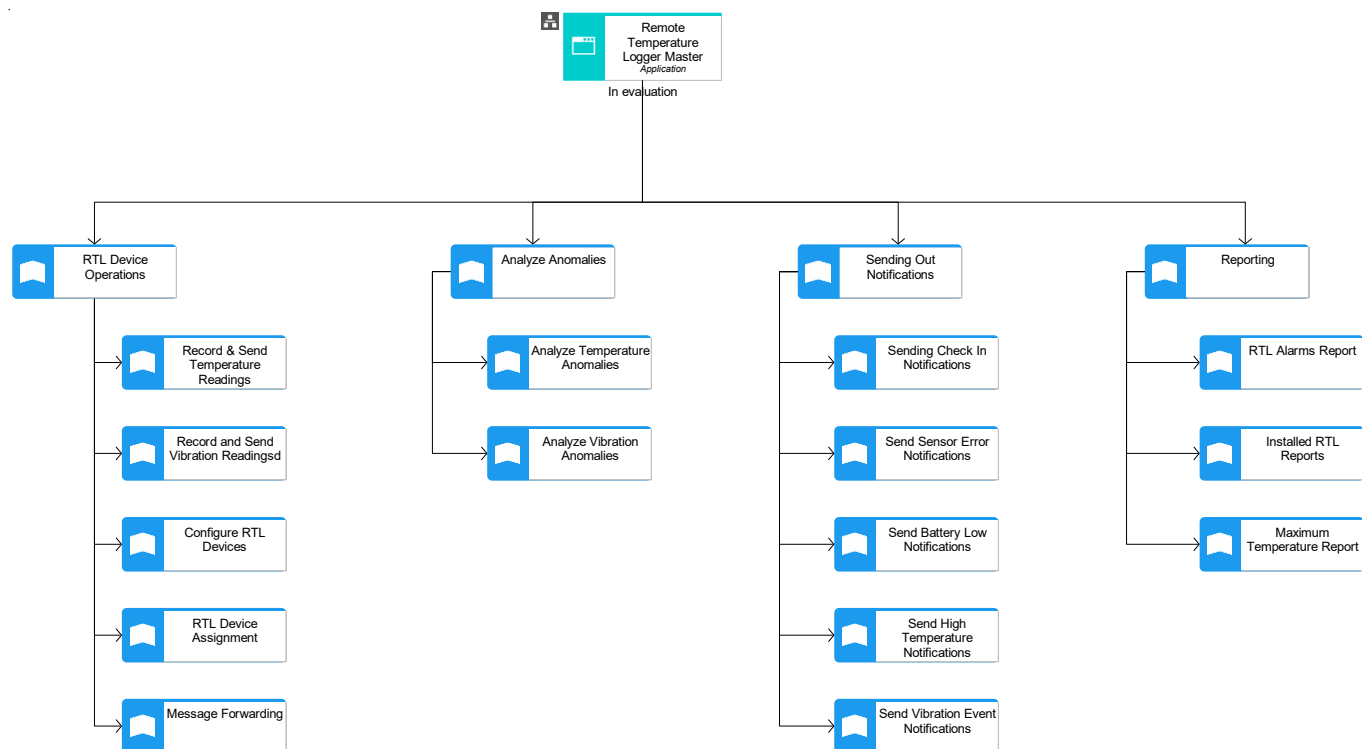
6.3 APPLICATION OVERVIEW

The RTL Application will be a web-based application that uses web technologies with supported communication protocols. The RTL Application will also make use of machine learning models for detection of anomalies in the recorded temperature and vibration readings. MQTT, which is an IOT messaging protocol will be used for communication between the RTL devices and the RTL Application. MQTT is an extremely lightweight publish/subscribe messaging transport that is ideal for connecting remote devices with minimal network bandwidth will be used

6.3.1 Use cases that are met by the application:

1. Monitoring and analysing transformer temperature and vibrations
2. Provide reporting of the recorded temperature and vibrations readings
3. Configuration of the RTL devices (IOT devices) via the RTL application

6.3.2 Application Functional Decomposition



6.4 APPLICATION STRATEGY AND PACKAGING

The RTL application will be internally developed as a Pilot solution under the Digital Incubator initiative of the Research Testing and Development. Outcome of the Pilot will determine whether further action should be taken to have the solution implemented in production for use by business.

7 INTEGRATION ARCHITECTURE

7.1 INTEGRATION INTERFACE MAP

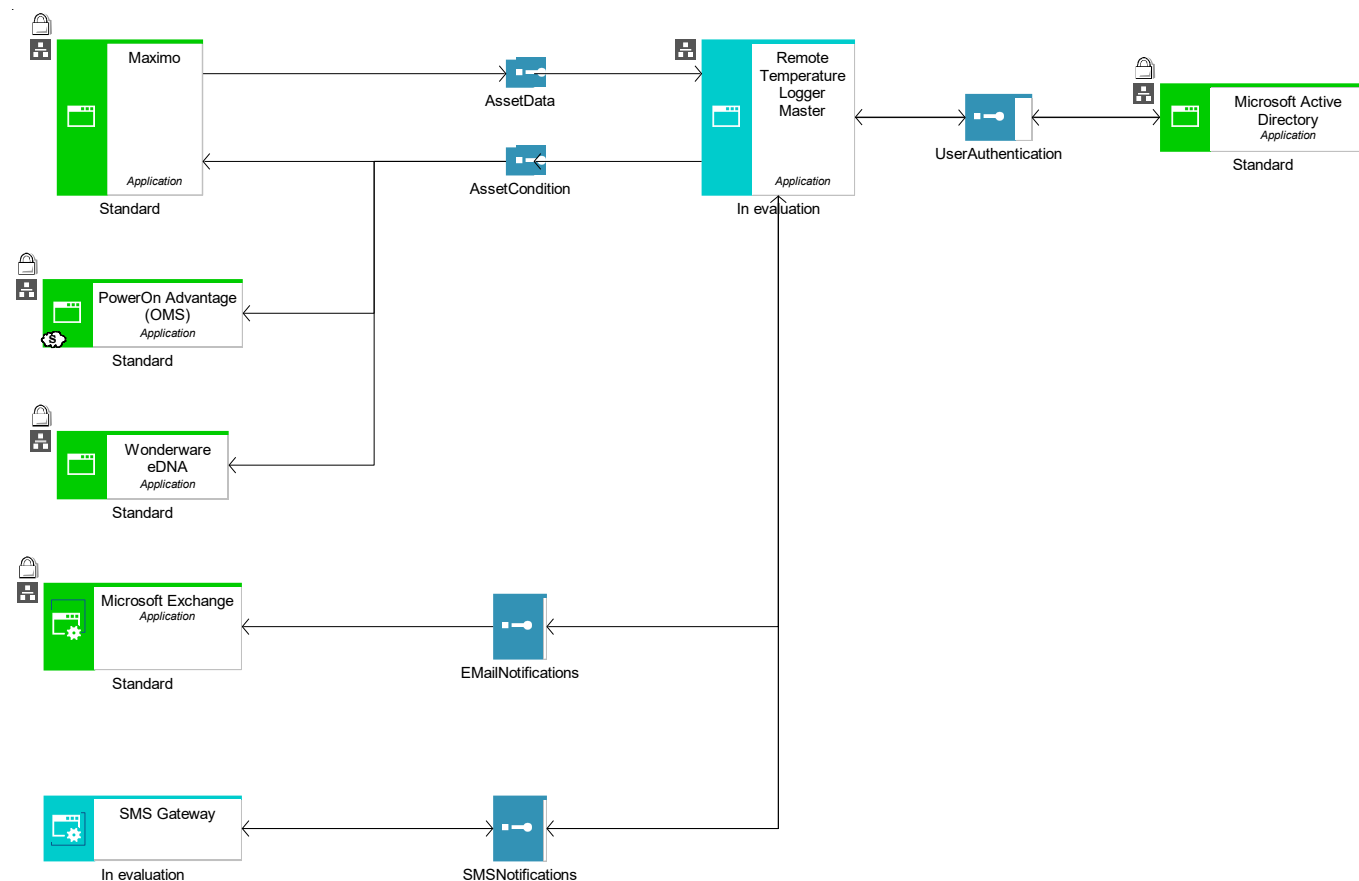


Figure 1 Application communication diagram

Interface Name	Source System	Destination System	Frequency	Description
AssetDetails	Maximo	RTL Application	Near Real Time	For updating transformer details to the RTL Application
AssetCondition	RTL Application	Maximo, Wonderware eDNA, PowerOn Advantage	Real Time	For updating transformer condition
EmailNotifications	RTL Application	Microsoft Exchange Server	Real Time	For sending out email notifications to users
SMSNotifications	RTL Application	SMS Gateway	Real Time	For sending out SMS notifications to users

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UserAuthentication	Microsoft Entra ID	RTL Application	Real Time	For user authentication
EmployeeDetails	SAP HR	RTL Application	Real Time	For retrieving Eskom employee details

AssetDetails Interface Fields Mapping

Field Name on RTL Application	Data Type	Mandatory	Integration Bus Field Transformations	Field on Maximo
TransformerID	varchar	Yes	TransformerID=AssetId	Location
OperatingUnit	varchar	Yes	OperatingUnit= OperatingUnit	
Zone	varchar	Yes	Zone=Zone	
Sector	varchar	Yes	Sector=Sector	
CNC	varchar	Yes	CNC=CNC	
FeederName	varchar	Yes	FeederName=FeederName	

AssetCondition Interface Fields Mapping

Field Name on RTL Application	Data Type	Mandatory	Integration Bus Field Transformations	Field on Maximo, Wonderware eDNA, PowerOn Advantage
TransformerID	varchar	Yes	TransformerID=AssetId	AssetId
OperatingUnit	varchar	No	OperatingUnit= OperatingUnit	
Zone	varchar	No	Zone=Zone	
Sector	varchar	No	Sector=Sector	
CNC	varchar	No	CNC=CNC	
FeederName	varchar	No	FeederName=FeederName	
TemperatureAlert	integer	No	TemperatureAlert=TemperatureAlert	
VibrationAlert	integer	No	VibrationAlert = VibrationAlert	

SMSNotification Interface Fields Mapping

Field Name on RTL Application	Data Type	Mandatory	Integration Bus Field Transformations	Field on SMS Gateway
MobileNumber	varchar	Yes	MobileNumber = MobileNumber	MobileNumber
UID	varchar	No	UID = UID	UID
TransformerID	varchar	No	TransformerID = TransformerID	TransformerID
MessageType	varchar	No	MessageType = MessageType	MessageType
BatteryVoltage	varchar	No	BatteryVoltage = BatteryVoltage	BatteryVoltage

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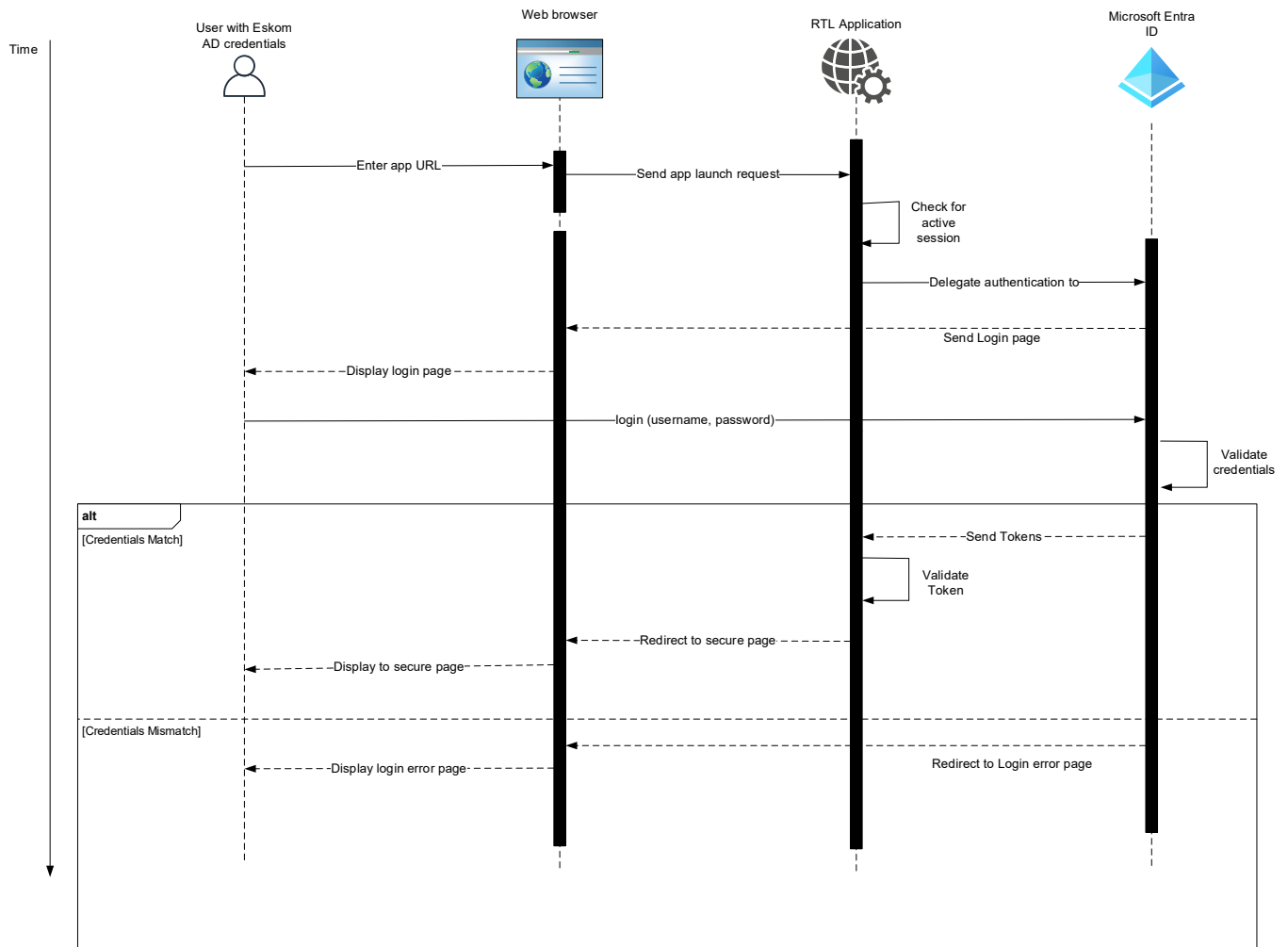
Temperature	integer	Yes	Temperature = Temperature	Temperature
Timestamp	integer	Yes	Vibration = Vibration	Vibration

EmployeeDetails Interface Fields Mapping

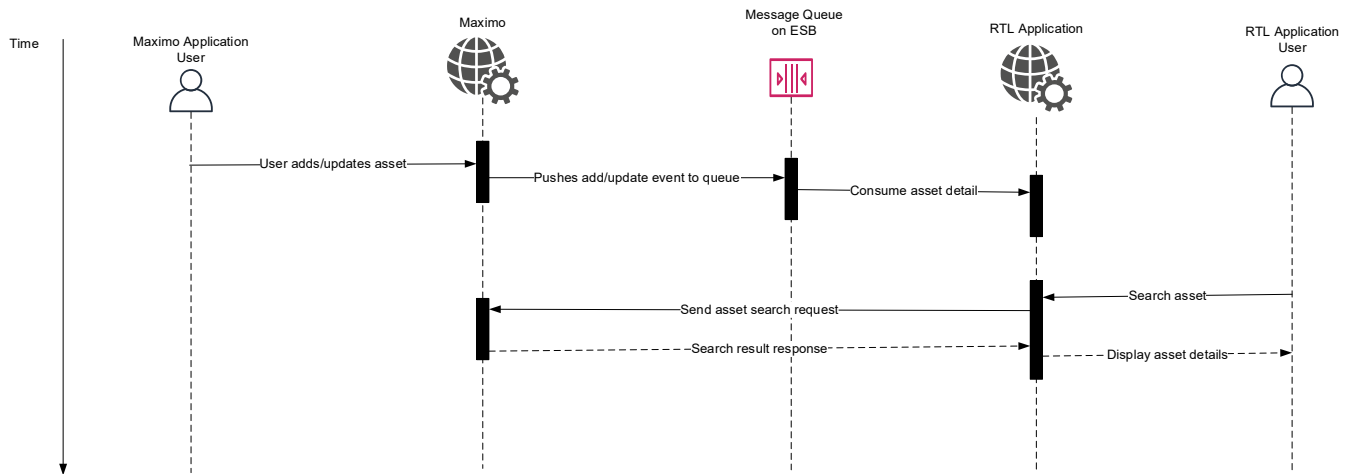
Field Name on RTL Application	Data Type	Mandatory	Integration Bus Field Transformations	Field on SMS Gateway
UniqueNumber	varchar	Yes	UniqueNumber = UniqueNumber	UniqueNumber
Name	varchar	No	Name = Name	Name
Surname	varchar	No	Surname = Surname	Surname
EmailAddress	varchar	No	EmailAddress = EmailAddress	EmailAddress
MobileNumber	varchar	No	MobileNumber = MobileNumber	MobileNumber
EndDate	integer	Yes	EndDate = EndDate	EndDate

7.2 INTERGRATION SEQUENCE DIAGRAMS

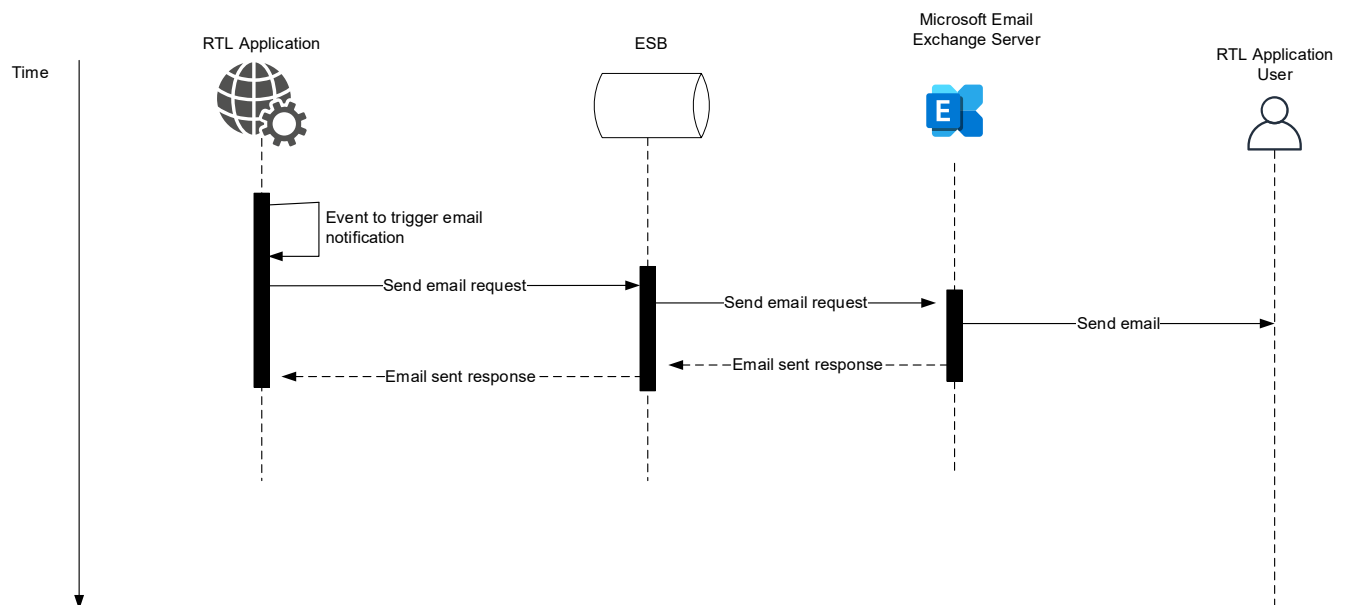
7.2.1 Authentication Sequence Diagram



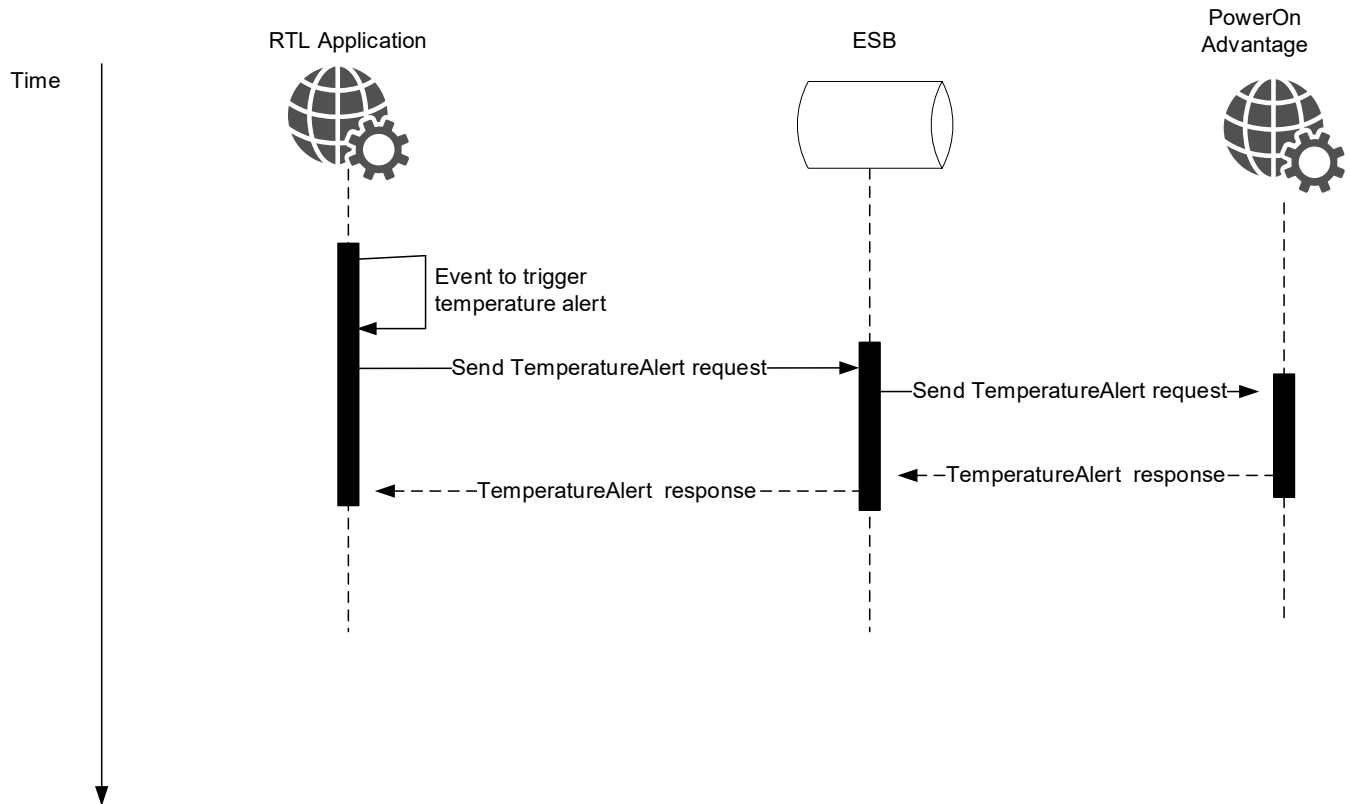
7.2.2 AssetDetails Sequence Diagram



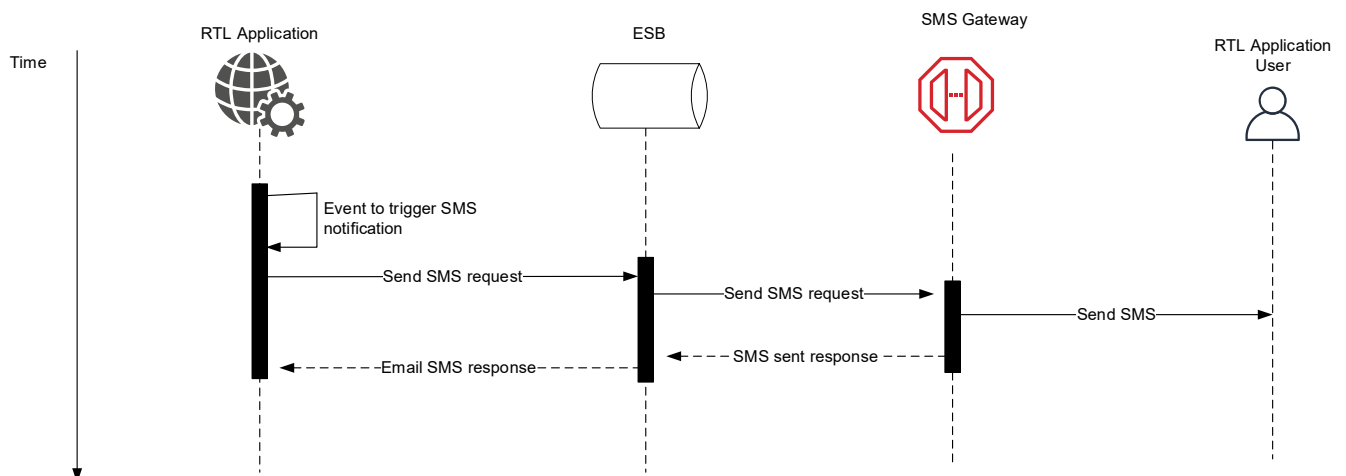
7.2.3 EmailNotifications Sequence Diagram



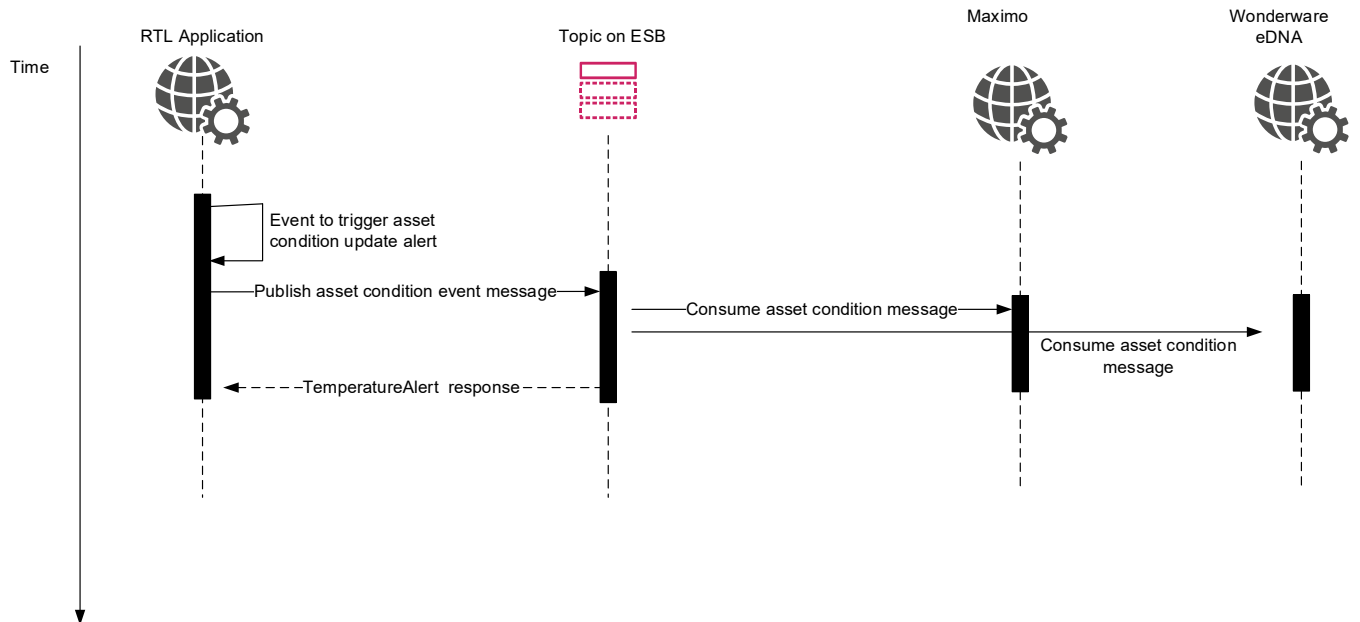
7.2.4 TemperatureAlert Sequence Diagram



7.2.5 SMSNotification Sequence Diagram



7.2.6 AssetCondition Sequence Diagram



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7.3 USER EXPERIENCE

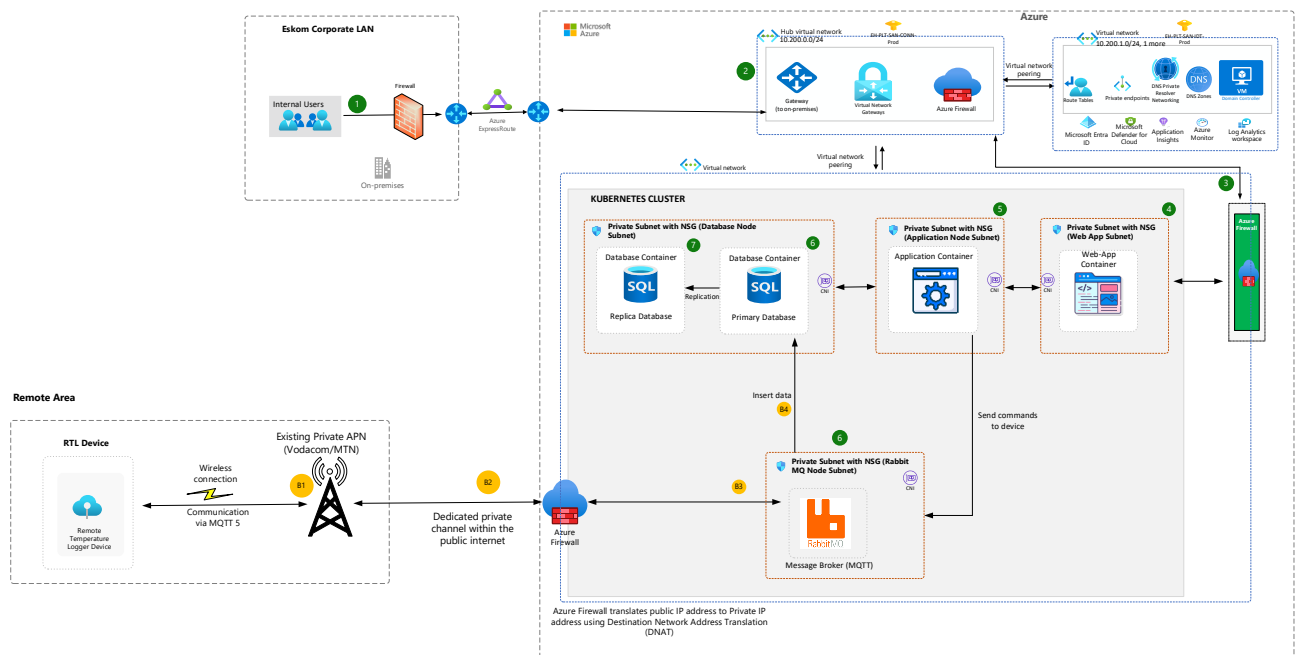
7.3.1 Proposed interface and its ease of use

The solution will use a web-based interface that can be accessed via Eskom approved browsers. The interface should be easy to navigate and provide users with clear error messages.

8 TECHNICAL ARCHITECTURE

8.1 Basic Infrastructure

The following diagram depicts the Design, the Services, the Communication Paths and the Basic Infrastructure:



Remote Temperature Logger Device

The Remote Temperature Logger Device (RTL Device) will consist of a microcontroller unit interfaced with temperature and vibration sensors. The RTL Device will run on the Real Time Operating System. Communication to the RTL Application will be over a wireless connection that uses either MTN or Vodacom Private APN.

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Stream Processing

The RTL device uses a private wireless APN to connect to the RTL Application. RTL devices will send data using MQTT protocol. The traffic goes through an Azure Firewall. The Azure Firewall will translate public IP address to Private IP address using Destination Network Address Translation (DNAT).

Messages from the RTL devices are published to a topic hosted on RabbitMQ (MQTT broker). RTL Application will consume messages from the topic on RabbitMQ and insert them into the databases.

RTL Application

The front-end is hosted in a Web-App container inside a Pod in a Kubernetes Cluster. The back-end is hosted in a separate Application Container inside another Pod. The database both primary and replica are hosted in separate containers and Pods. The containers for the front-end, back-end and databases are in different Pods to allow independent scaling, management and deployment. Kubernetes Service will be used to enable communication between the Pods using stable DNS names.

8.1.1 Infrastructure

8.1.2 Application server

Table 8.1 List Of Application Servers

New / Upgrade / Replaced	Previous server	Manufacturer	Model	Server Name	Primary Applications(s)	Operating System	IP Addresses	Site	Tier	Total No. Of vCPU's	CPU Clock Speed	Total Shared Memory	Total Storage
TBD													

8.1.3 Database Server

Table 8.2 List of Database Servers

New / Upgrade / Replaced	Previous server	Manufacturer	Model	Server Name	Primary Applications(s)	Operating System	IP Addresses	Site	Tier	Total No. Of vCPU's	CPU Clock Speed	Total Shared Memory	Total Storage
TBD													

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8.1.4 Web Server

Table 8.3 List of Web Servers

3New / Upgrade d / Replace d	Previous server	Manufacturer	Model	Server Name	Primary Applications(s)	Operating System	IP Addresses	Site	Tier	Total No. Of vCPU's	CPU Clock Speed	Total Shared Memory	Total Storage
TBD													

8.2 CLOUD

8.2.1 Services provided by Cloud

Below mentioned services will be offered by the cloud:

1. Virtual Machines
2. Azure SQL Database (managed instance)
3. Virtual Network
4. Azure Data Explorer

8.2.2 Cloud to Eskom IT systems interfaces

Azure ExpressRoute will be used to extend Eskom's on-premises network into the Microsoft cloud over a private connection with the help of a connectivity provider (Teraco).

8.3 NETWORK AND TELECOMS

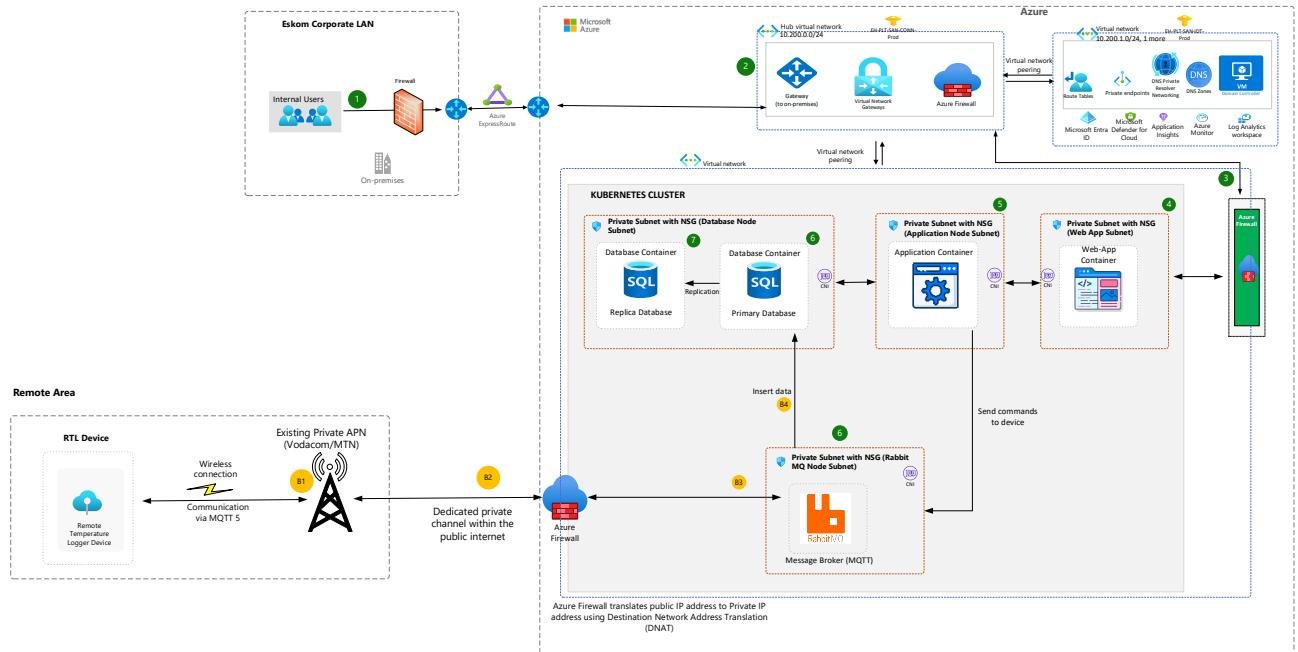
8.3.1 Applied Technologies

The existing deployment technologies that will be applied during this deployment include the following:

1. Network firewalls (On-prem)
2. Virtual Machines
3. Azure firewalls
4. Azure ExpressRoute private connection

8.4 PHYSICAL DESIGN

8.4.1 Physical Design as depicted below.



8.5 USER BASE

8.5.1 User Base

The solution will have a total of 15 users during the Pilot project. The total number of concurrent users will be 15.

8.6 SOFTWARE AND REQUIREMENTS

8.6.1 Servers (Software)) (DB/Web/App etc.), Platform, OS, Hardware and Virtualisation

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Server Name	Software Category	Software Package A and its exact version number	Software Package A
Eskom Azure Tenant TBD	Eskom Azure Tenant TBD	TBD	TBD

8.7 LICENCING

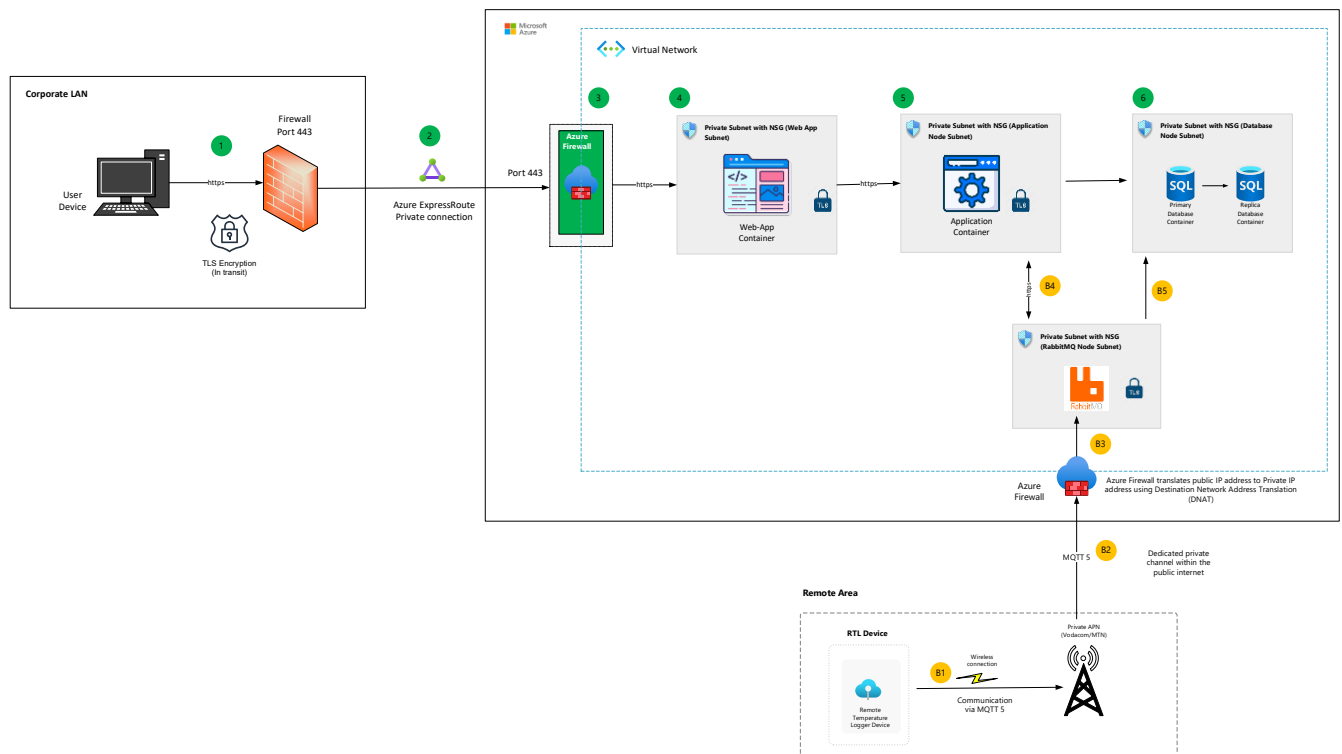
The solution will be hosted on Microsoft Azure Cloud using Eskom's tenant. The current credit subscription model in use by Eskom will be adopted for the provisioning of all the services required for the RTL Application.

8.8 VERSION CONTROL

Version control of the implemented solution will be performed using CI/CD methodology using the Azure DevOps tool.

9 SECURITY ARCHITECTURE

9.1 SECURITY ARCHITECTURE



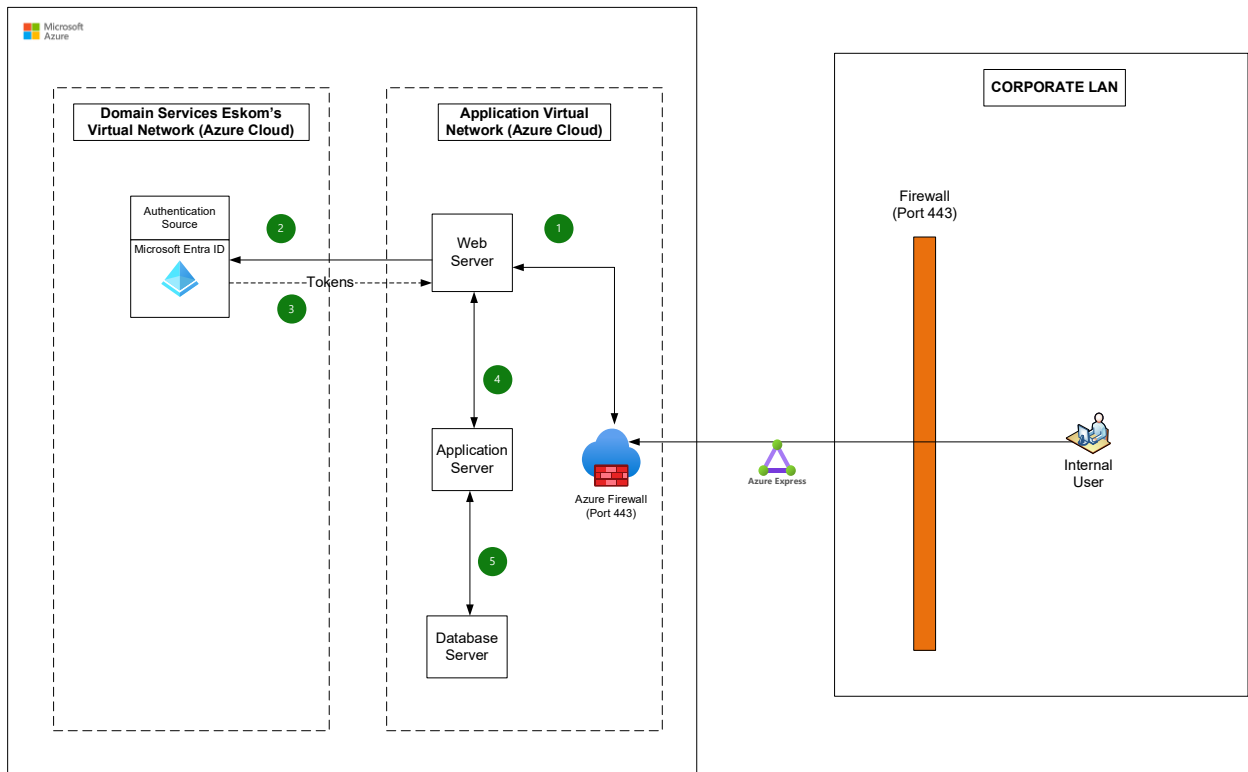
1. End user devices located in Eskom's Corporate LAN communicates with the RTL application using HTTPS (data in transit encryption).
2. Azure ExpressRoute will provide a private connection between Eskom Corporate LAN and Azure Cloud.
3. Firewalls on the Corporate LAN and Azure Firewall on the Azure Cloud will filter specific types of traffic
4. A Kubernetes cluster hosted inside a Virtual Network with Web-App container, Application container and database containers each hosted in a Private subnet with Network Security Group (NSG) configured with rules to filter traffic that flow into and out of the each subnet. The network security group will have rules to deny or allow traffic to the subnets hosting the web front-end, back-end and databases.

B1. The Remote Temperature Logger device (which is a microcontroller unit interfaced with temperature and vibration sensors) will have security controls configured on it. Below are the security controls that will be configured on the RTL Device:

- Secure Bootloader – to prevent unauthorized upload of programs to the microcontroller unit
- Physical Tampering Resistance – use of tamper-resistant hardware to resist physical attacks, such as drilling, cutting, or prying, and to trigger self-destruction or erasure mechanisms if tampered with
- Encrypting data at rest and data in transit
- Updating the devices by installing the latest patches, updates, and security fixes for the firmware, software and applications

B2. Data from the devices will be transmitted wirelessly using MTN and Vodacom Eskom's Private APNs. The data will be transmitted using HTTPS from the RTL Devices.

9.2 USER IDENTIFICATION AND AUTHENTICATION



1. Internal user enters application URL on a browser and the request goes through the Azure ExpressRoute private connection to the web server in the Web Server.

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2. Web server redirects the user to the Microsoft Entra ID login page where the user submits AD credentials.
3. If the user credentials are successfully validated, a response message with an authentication and refresh token will be sent back to the web server. If the user credentials are not successfully validated, a login failed response message will be returned to the user. The web server validates the received tokens thereafter returns a secure homepage to the user.
- 4&5. After successful authentication user request can then be forwarded to the back end of the application for processing

9.3 SECURITY GUARD RAILS

The RTL Application must comply to Eskom's IT Security Standards namely:

1. 32-351 Information Security Logical Access Standard Revision 5
2. 240-107007584 Web Services Security Standard Revision 3
3. 240-107007512 Database Security Standard Revision 4
4. 240-97536243 Integration Security Standard Revision 4

9.4 AUDIT TRAILS

Audit trail of system user activities will be recorded and stored with details such as timestamp, user, operation, old and new values. Audit trail data to be stored both on the RTL Application database in the audit table.

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10 APPENDIX A (DEFINITIONS)

10.1 DEFINITIONS

10.1.1 Architecture Specific Definitions

10.1.1.1 *Architecture*

A formal description of a system, or a detailed plan of the system at component level, to guide its implementation. The structure of components, their inter-relationships, and the principles and guidelines governing their design and evolution over time (TOGAF).

10.1.1.2 *Architecture Vision*

A succinct description of the Target Architecture that describes its business value and the changes to the enterprise that will result from its successful deployment. It serves as an aspirational vision and a boundary for detailed architecture development (TOGAF).

10.1.1.3 *Architecture Development Method (ADM)*

A step-by-step approach to develop and use the enterprise architecture (TOGAF).

10.1.1.4 *Baseline Architecture*

A specification that has been formally reviewed and agreed upon, that thereafter serves as the basis for further development or change and that can be changed only through formal change control procedures or a type of procedure such as configuration management (TOGAF).

10.1.1.5 *Integration*

This is the linking of multiple application systems and their subsystems to communicate together seamlessly.

10.1.1.6 *Target Architecture*

The description of a future state of the architecture being developed for an organization. There may be several future states developed as a roadmap to show the evolution of the architecture to a target state (TOGAF).

10.1.1.7 *Work Package*

A logical group of architecture development work identified to achieve one or more objectives for the business. A work package can be created as result of a business request for IT work, an approved change request, terms of reference for architecture work or from migration planning (TOGAF)

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10.1.2 Other Definitions

10.1.2.1 Big Data

Big data is high-volume, -velocity and –variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making (Smith, David M., 2012).

10.1.2.2 Data Centre

A data centre is a special building constructed to house servers. The data centre consists of a number of rooms with special filtration and air conditioning to keep servers free of dust and at their optimum operating temperature and preventing the significant quantities of heat they generate from building up and damaging the equipment. The building provides special electricity supplies, with generators that supply power in the event of a power cut and provide physical security in the form of guards and locks to ensure that only the owners of the servers and their appointed staff can gain access to a server, to prevent theft. Lastly, data centres have multiple high-quality Internet connections with backups, allowing the servers within to remain connected to the Internet at high speed, even if faults or accidents break one or more of the connections.

10.1.2.3 Disaster Recovery

Disaster recovery is the process, policies and procedures related to preparing for recovery or continuation of technology infrastructure critical to an organisation after a natural or human-induced disaster. Disaster recovery is a subset of business continuity. While business continuity involves planning for keeping all aspects of a business functioning in the midst of disruptive events, disaster recovery focuses on the IT or technology systems that support business functions.

10.1.2.4 Gartner Hype Cycle

*A **Hype Cycle** is a graphical depiction of a common pattern that arises with each new technology or other innovation. A technology (or related innovation) passes through several stages (technology trigger, peak of inflated expectations, trough of disillusionment, slope of enlightenment, and plateau of productivity) on its path to productivity:*

Technology Trigger: *The Hype Cycle starts when a breakthrough, public demonstration, product launch, or some other event generates press and industry interest in a technology innovation.*

Peak of Inflated Expectations: *A wave of “buzz” builds and the expectations for this new technology rise above the current reality of its capabilities. In some cases an investment bubble forms, as happened with the Web, social media and cloud computing.*

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Trough of Disillusionment: Inevitably, impatience for results begins to replace the original excitement about potential value. Problems with performance, slower-than-expected adoption or a failure to deliver financial returns in the time anticipated all lead to missed expectations, and disillusionment sets in.

Slope of Enlightenment: Some early adopters overcome the initial hurdles, begin to experience benefits and recommit efforts to move forward. Drawing on the experience of the early adopters, understanding grows about where and how the technology can be used to good effect and, just as importantly, where it brings little or no value.

Plateau of Productivity: With the real-world benefits of the technology demonstrated and accepted, growing numbers of organizations feel comfortable with the now greatly reduced levels of risk. A sharp uptick (“hockey stick”) in adoption begins, and penetration accelerates rapidly as a result of productive and useful value.

10.1.2.5 Real-Time Infrastructure

Real-Time Infrastructure (RTI) represents a shared IT infrastructure in which business policies and SLAs drive the dynamic allocation and optimization of IT resources, so that service levels are predictable and consistent, despite the unpredictable IT service demand. RTI provides the elasticity functionality and dynamic optimization and tuning of the runtime environment based on policies and priorities across private, public and hybrid cloud architectures. Where resources are constrained, business policies determine how resources are allocated to meet business goals

10.1.2.6 Server Room

A server room has the same function as a data centre but is much smaller and does not require the same level of availability as with a data centre.

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11 APPENDIX B (ABBREVIATIONS)

11.1 GENERAL GLOSSARY OF ABBREVIATIONS

Table 11.1 Glossary Of Abbreviations

Abbreviation	Description
ACF	Architecture Content Framework
ADM	Architecture Development Method
API	Application Programming Interface
BAM	Business Activity Monitoring
BC	Business Critical
Bpass	Business Process as a Service
BMS	Building Management System
BPEL	Business Process Execution Language
BPM	Business Process Management
CAD	Conceptual Architecture Definition
CAPEX	Capital Expenditure
CBS	Cloud Services Brokerage
CoE	Centre of Excellence
CFD	Computational Fluid Dynamic
CIM	Common Information Model
CIO	Chief Information Officer
COTS	Commercial off the Shelf
CN	Core Network
CRA	Concept Release Approval
DC	Data Centre
dbPaas	Database Platform as a Service
DMZ	Demilitarized Zone
DPRP	Disaster Prevention and Recovery Plan
DR	Disaster Recovery

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DRA	Design Release Approval
EAI	Enterprise Application Integration
EHPUM	Eskom High Performance Utility Model
ESB	Enterprise Service Bus
ERA	Execution Release Approval
EUC	End User Computing
FBI	Fabric-Based Infrastructure
FIFO	First In, First Out
FTP	File Transfer Protocol
HTTP	Hypertext Transfer Protocol
IaaS	Infrastructure as a Service
IA	Information Assurance
ICoE	Integration Centre of Excellence
ICT	Information Communication Technology
IDE	Integrated Development Environment
IEP	Intelligent Event Processing
IRA	Implementation Release Approval
IS	Information System
IT	Information Technology
JavaCAPS	Java Composite Application Platform Suite
JB1	Java Business Integration
JMS	Java Message Service
LAD	Logical Architecture Definition
LAN	Local Area Network
LDAP	Lightweight Directory Access Protocol
MC	Mission Critical
MDM	Master Data Management
NC	Non-Critical Systems
NIST	National Institute of Standards and Technology
OPEX	Operating Expenses
OT	Operational Technology

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PaaS	Platform as a Service
PAD	Physical Architecture Definition
POC	Proof of Concept
POJO	Plain Old Java Object
QA	Quality Assurance
RTI	Real Time Infrastructure
RTO	Return to Operation
SaaS	Software as a Service
SBB	Solution Building Blocks
SC	Safety Critical
SLA	Service Level Agreement
SMS	Short Message Service
SOA	Service-Oriented Architecture
SoAW	Statement of Architecture Work
SOP	Standard Operating Procedure
SQL	Structured Query Language
STCMS	SeeBeyond Technology Corporation Message Service
TBC	To Be Confirmed
TCoE	Testing Centre of Excellence
TCP/IP	Transmission Control Protocol and Internet Protocol
TOGAF	The Open Group Architecture Framework
UI	User Interface
UML	Unified Modelling Language
UNIX	Uniplexed Information Computing System
UPS	Uninterruptible Power Supplies
UX	User Interface
WP	Work Package
WSDL	Web Services Description Language
XML	Extensible Markup Language
XSD	XML Schema Definition
XSLT	Extensible Stylesheet Language Transformations

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12 ESKOM DOCUMENTATION (STANDARDS COMMONLY USED)

Document Number	Version	Title
32-389	1.12	Information security- Removable media standard.
32-385	1.14	IT Continuity/DR Standard
32-371	3.1	Group IT Change Control Standard
32-368	1.1	Information security: Incident management standard
32-369	2.4	Information security: Physical asset classification standard
32-351	2.8	Information security- Logical access control standard
32-723	1.1	Backup and Recovery Standard
240-44071782	1.11	Information Security Oracle Baseline 11g
240-86823222	1.7	Group IT: Eskom Linux 7 Technical Security Standard
240-87121701	1.6	Group IT: MySQL 4.1,5&5.1 Technical Security Standard
240-87121845	1.5	Group IT: Oracle Solaris 11 technical Security Standard
240-87121993	1.6	Group IT: Oracle Solaris 10 technical Security Standard
240-86966307	1.5	Group IT: Microsoft Exchange 2013 Technical Security Standard
240-86966667	1.5	Group IT: Microsoft IIS 7.0 Technical Security Standard
240-86966237	1.5	Group IT: Microsoft Server 2008 R2 Technical Security Standard
240-86966811	1.5	Group IT: VMWare ESXi 5.1 Technical Security Standard
240-87430851	1.5	Group IT: IBM AIX 7.0 Technical Security Standard
240-89612923	1.4	Group IT: Apache HTTP Server 2.2 Technical Security Standard
240-89613371	1.4	Group IT: Cisco Wireless LAN Controller 7 Technical Security Standard
240-90006778	1.4	Group IT: MS SQL Server 2000 Technical Security Standard
240-90041165	1.4	Group IT: MS SQL Server 2005 Technical Security Standard
240-90045717	1.4	Group IT: Apache Tomcat 5.5/6.0 Technical Security Standard
240-86830408	1.6	Group IT: Eskom Apache HTTP Server 2.4 Technical Security Standard
240-86832077	1.7	Group IT: Eskom Windows Server 2012 Technical Security Standard
240-89613769	1.4	Group IT: VMware ESX 3.5 Technical Security Standard
240-90040591	1.4	Group IT: VMware ESX 4 Technical Security Standard
240-95460364	1.3	Group IT: Eskom IIS 5&6 Technical Security Standard
240-96587315	1.4	Group IT: Oracle 12 database Technical Security Standard
240-102337621	1.2	Group IT: Oracle 8i Technical Security Standard

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240-102337689	1.2	Group IT: Oracle Linux 5 Technical Security Standard
240-102338299	1.2	Group IT: SUSE 9 Linux Technical Security Standard
240-102337250	1.2	Group IT: AIX 6.1 Technical Security Standard
240-102337440	1.2	Group IT: AIX 5.3 Technical Security Standard
240-102337524	1.2	Group IT: SUSE Linux Enterprise Server 11 Technical Security Standard
240-102337763	1.2	Group IT: Microsoft Windows Sever 2003 Technical Security Standard
240-102337811	1.2	Group IT: Red Hat Enterprise Linux 5 Technical Security Standard
240-102338007	1.3	Group IT: SUSE 10 Linux Technical Security Standard
32-373	4.4	Information security - IT/OT and third-party remote access standard (Supersedes ESKASADE4)
240-103865868	1.3	Group IT: Windows 10 Technical Security Standard
32-377	1.16	Information Security - Firewall Standard
240-103865834	1.4	Group IT: Window 8.1 Technical Security Standard
240-97536243	1	Integration Security Standard
240-106727489	1.2	Information Security email archiving Standard
240-107007512	1.1	Database Security Standard
240-107007584	1.1	Web Services Security Standard
206-1461		Recipe - Statement of Architecture Work
206-1462 Recipe		Conceptual Architecture Definition of an Application
206-1513		Logical Architecture Definition of an Application
206-1630		Logical Architecture Definition of a Data Interface
206-1241		TOGAF@E Architecture Content Framework
206-1443		TOGAF@E Content Metamodel